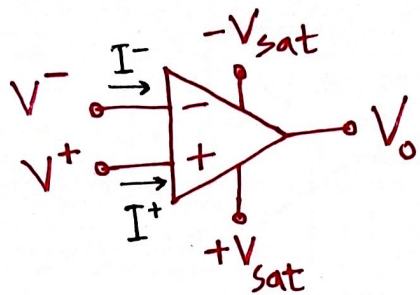


Op-Amp



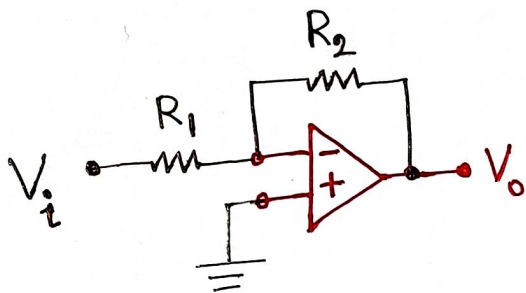
here, $V^- \rightarrow$ inverting input
 $V^+ \rightarrow$ non-inverting input

$V_o \rightarrow$ output

$$V_o = \begin{cases} +V_{sat} & \text{if } V^+ > V^- \\ -V_{sat} & \text{if } V^- > V^+ \end{cases}$$

$$I^+ = I^- = 0$$

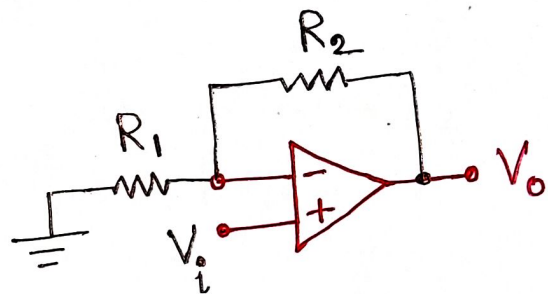
Inverting Amplifier



$$V_o = -\frac{R_2}{R_1} V_i$$

$$\text{here, Gain} = \frac{V_o}{V_i} = -\frac{R_2}{R_1}$$

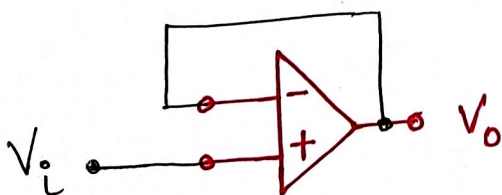
Non-Inverting Amplifier



$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_i$$

$$\text{here, Gain, } \frac{V_o}{V_i} = \left(1 + \frac{R_2}{R_1}\right)$$

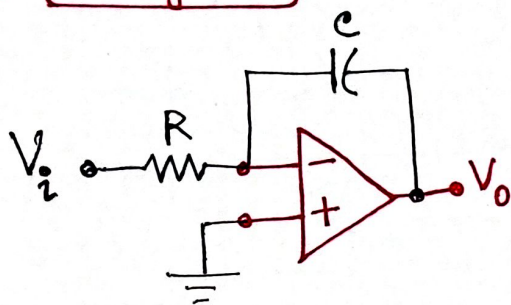
Buffer/Voltage Follower



$$V_o = V_i$$

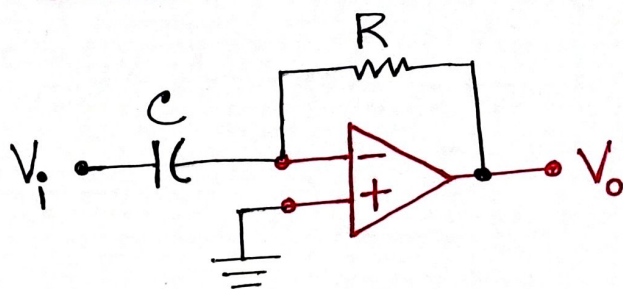
$$\text{here, Gain} = \frac{V_o}{V_i} = 1.$$

Integrator



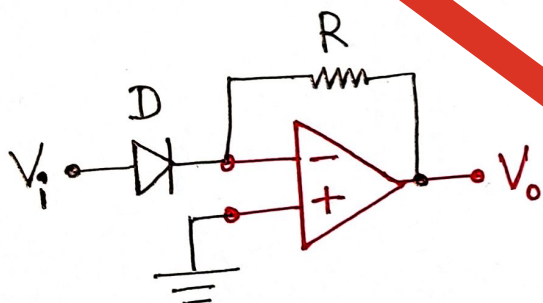
$$V_o = -\frac{1}{RC} \int V_i dt$$

Differentiator



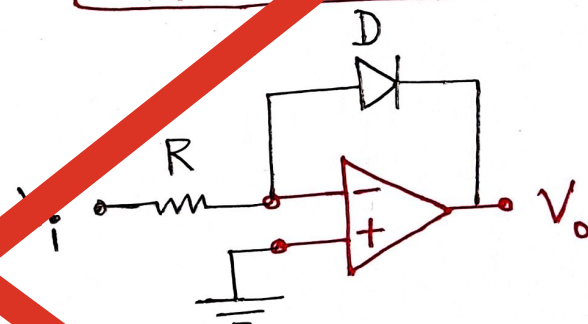
$$V_o = -RC \frac{d}{dt}(V_i)$$

Exponential Amplifier



$$V_o = -I_s R \exp\left(\frac{V_i}{V_T}\right)$$

Logarithmic Amplifier

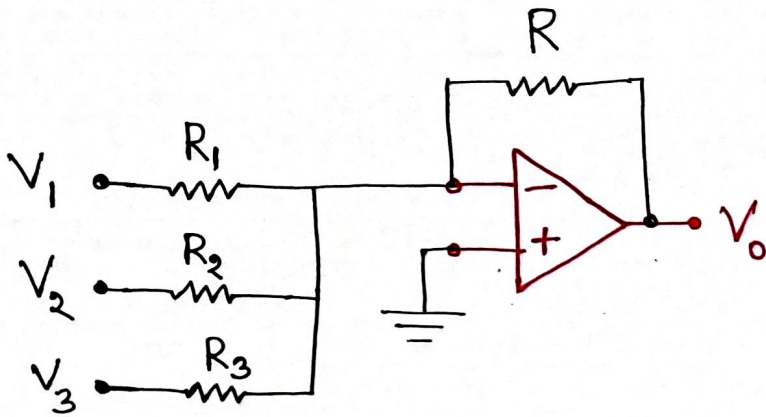


$$V_o = V_T \ln\left(\frac{V_i}{I_s R}\right)$$

here, $V_T = 25 \text{ mV}$ (at room temperature)

$$I_s = 10^{-9} \text{ A}$$

Inverting Adder



$$V_o = -\left(\frac{R}{R_1} V_1 + \frac{R}{R_2} V_2 + \frac{R}{R_3} V_3\right)$$

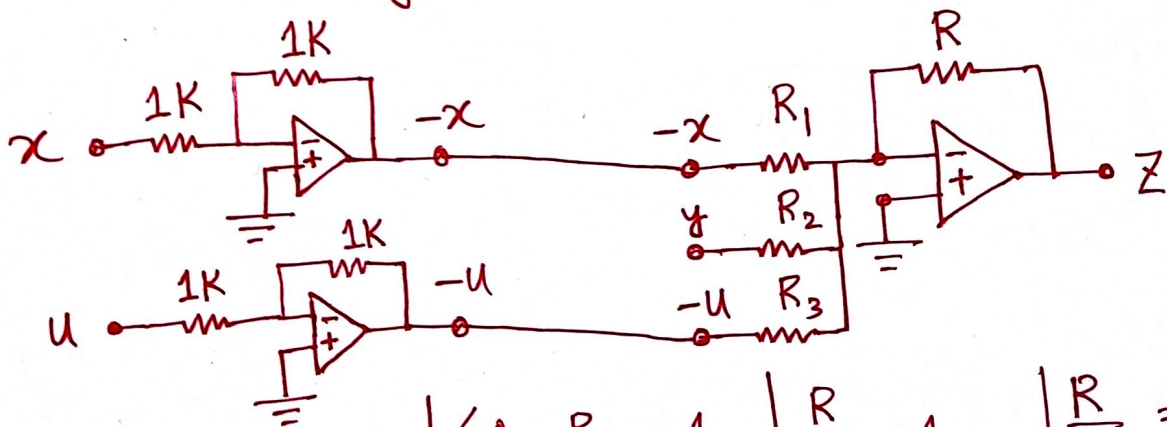
Examples

Implement the following mathematical operations using Op-Amp.

(i) $z = x - y + 5u$

Solⁿ

$$z = -(-x + y - 5u) = -((1)(-x) + (1)(y) + (5)(-u))$$



Let, $R = 10K$.

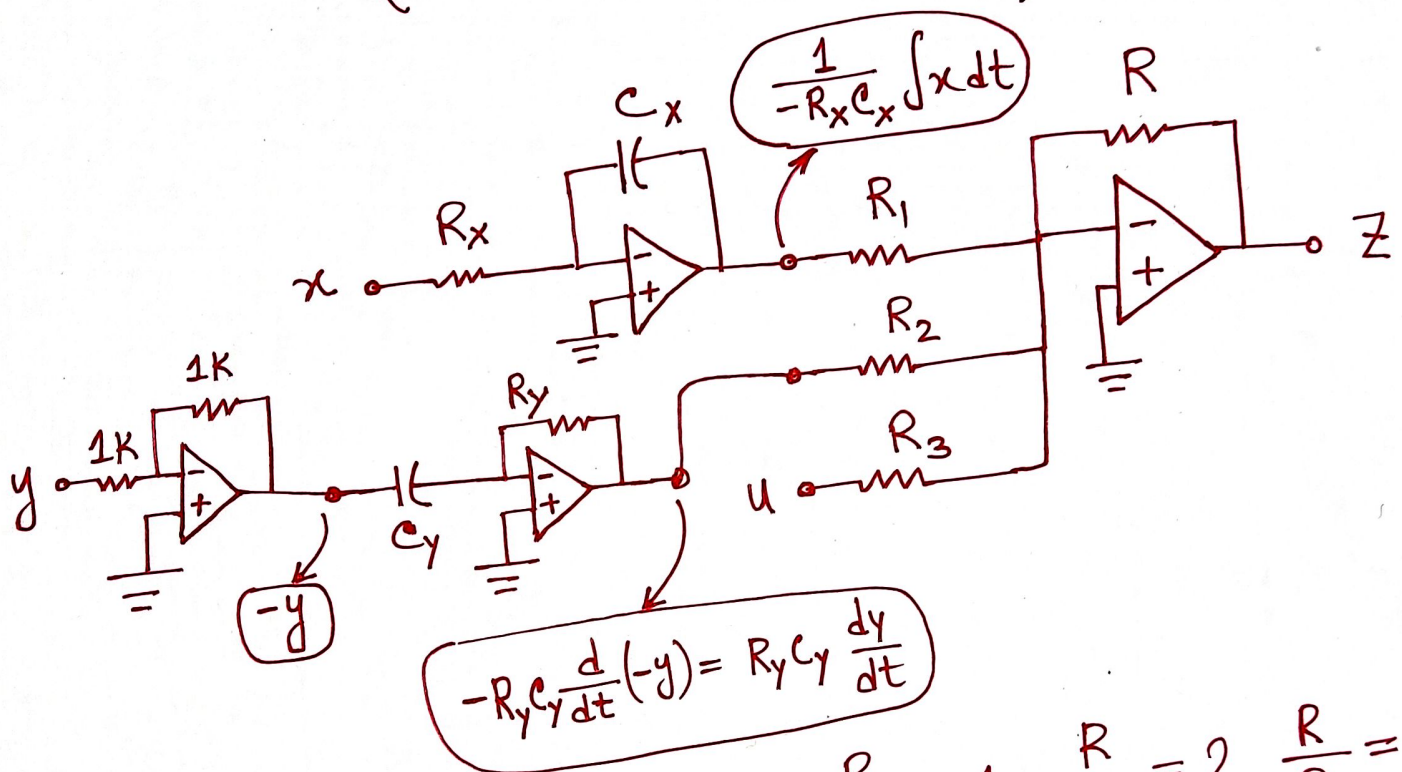
$\therefore, \frac{R}{R_1} = 1$	$\frac{R}{R_2} = 1$	$\frac{R}{R_3} = 5$
$\hookrightarrow R_1 = 10K$	$\hookrightarrow R_2 = 10K$	$\hookrightarrow R_3 = 2K$

$$\textcircled{2} \quad z = \int x \, dt - 2 \frac{dy}{dt} - u$$

Soln

$$z = - \left(- \int x \, dt + 2 \frac{dy}{dt} + u \right)$$

$$= - \left((1) \left(- \int x \, dt \right) + 2 \left(- \frac{dy}{dt} \right) (-1) + (1)(u) \right)$$



here,

$$\frac{-1}{R_x C_x} = -1$$

$$\text{let, } C_x = 47 \mu\text{F}$$

$$\therefore R_x = 21.2766 \text{ K}\Omega$$

$$\text{Again, } -R_y C_y = -1$$

$$\text{let, } C_y = 47 \mu\text{F}$$

$$\therefore R_y = 21.2766 \text{ K}\Omega$$

$$\text{Again, } \frac{R}{R_1} = 1, \frac{R}{R_2} = 2, \frac{R}{R_3} = 1$$

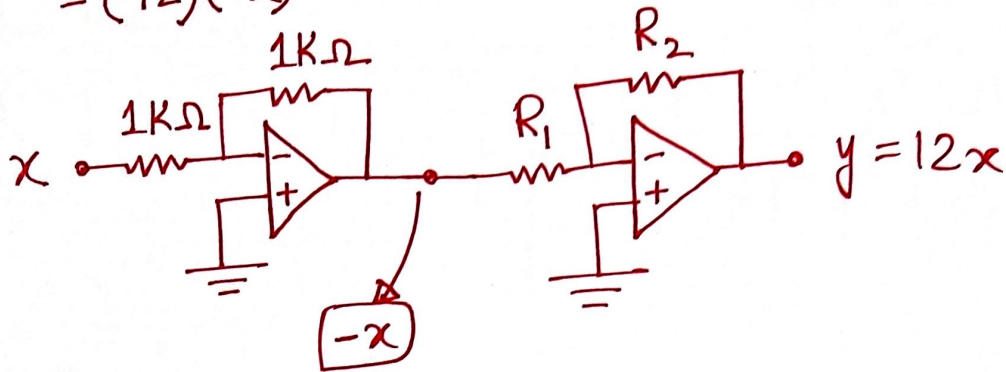
$$\text{let, } R = 10 \text{ K}\Omega$$

$$\therefore R_1 = 10 \text{ K}\Omega$$

$$R_2 = 5 \text{ K}\Omega$$

$$R_3 = 10 \text{ K}\Omega$$

③ $y = 12x$
 $= (-12)(-x)$



here, $-\frac{R_2}{R_1} = -12$

let, $R_1 = 1K\Omega$

$\therefore R_2 = 12K\Omega$

④ $y = x^2$

Solⁿ

$y = x^2$

$\hookrightarrow \ln y = \ln x^2$

$\hookrightarrow \ln y = \ln x + \ln x$

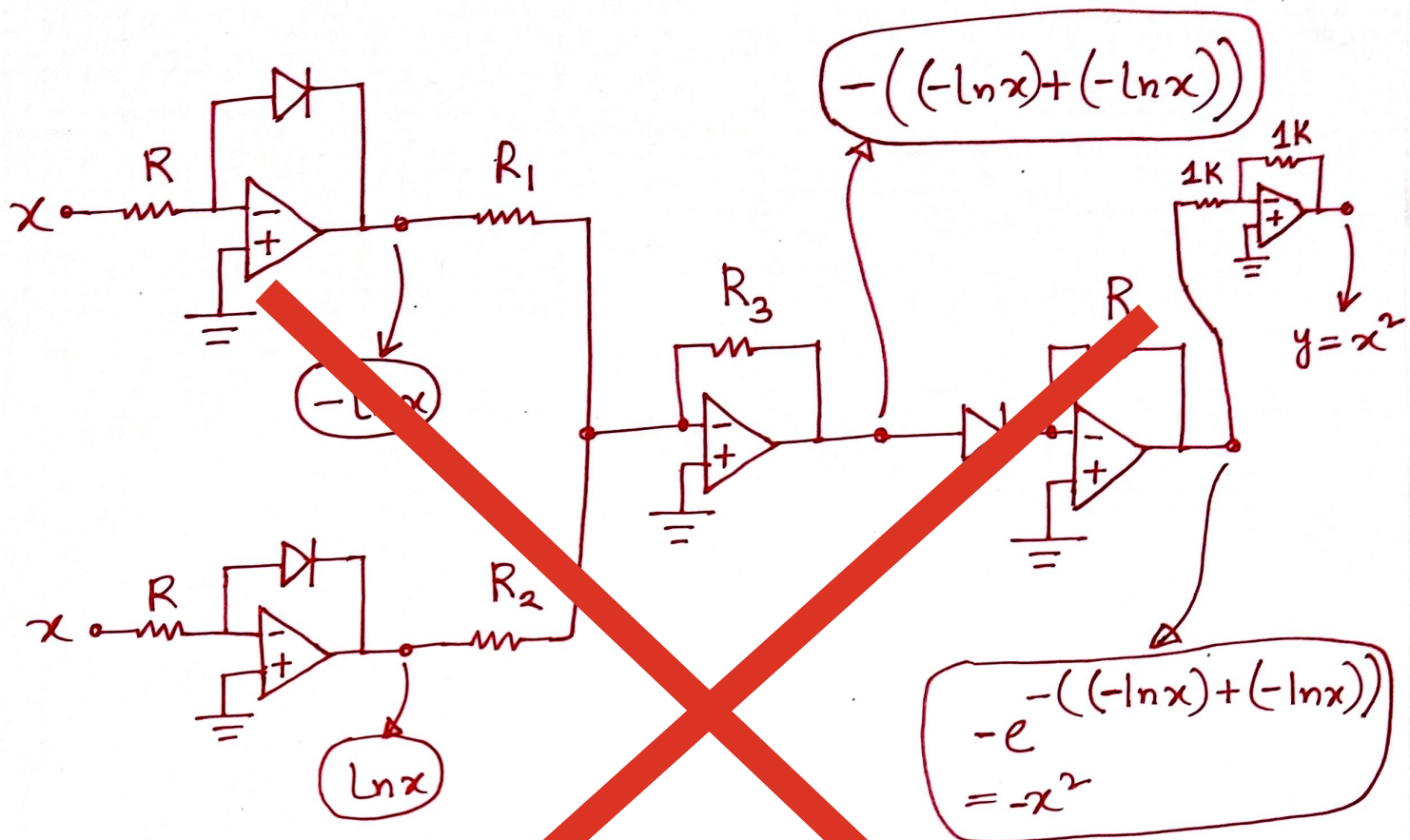
$\hookrightarrow y = e^{\ln x + \ln x}$

$\hookrightarrow y = e^{-((- \ln x) + (- \ln x))}$

we need to implement this function using exponential & logarithmic amplifier.

let, $I_S R = 1$

$V_T = 1.$

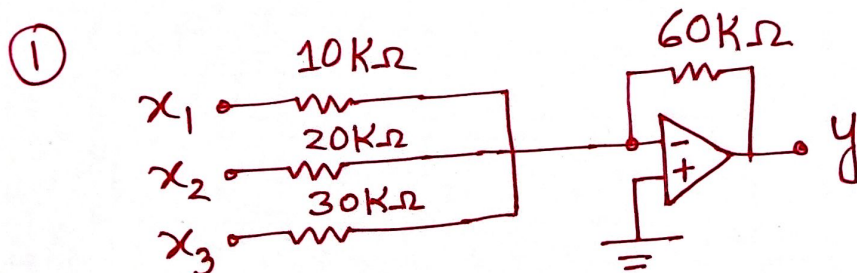


here, $\frac{R_3}{R_1} = 1$, $\frac{R_3}{R_2} = 1$

let, $R_3 = 10 \text{ K}\Omega$

$\therefore R_1 = 10 \text{ K}\Omega$ and $R_2 = 10 \text{ K}\Omega$

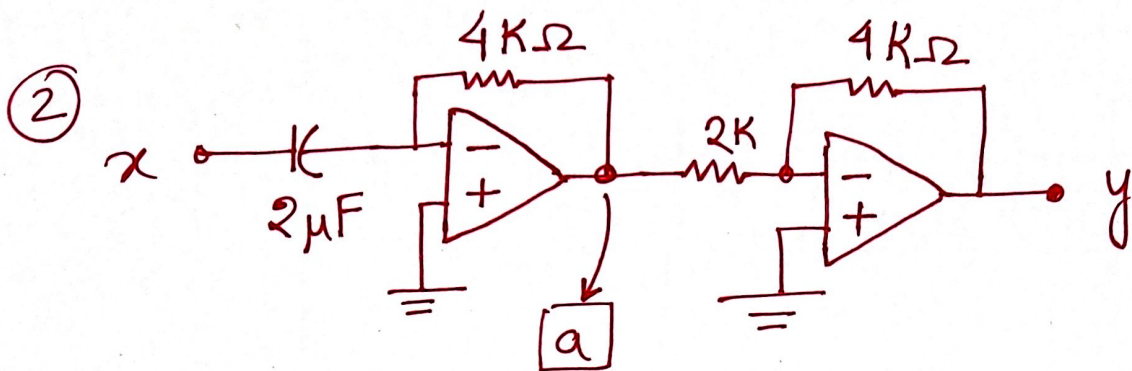
Find the output of the following circuits.



Solⁿ

$$y = -\left(\frac{60 \text{ K}\Omega}{10 \text{ K}\Omega} x_1 + \frac{60 \text{ K}\Omega}{20 \text{ K}\Omega} x_2 + \frac{60 \text{ K}\Omega}{30 \text{ K}\Omega} x_3\right)$$

$$\hookrightarrow y = -(6x_1 + 3x_2 + 2x_3)$$



$$a = -RC \frac{d}{dt}(x) = -2 \times 10^{-6} \times 4 \times 10^3 \times \frac{d}{dt}(x)$$

$$\hookrightarrow a = -8 \times 10^{-3} \times \frac{dx}{dt}$$

$$\therefore y = -\left(\frac{4k\Omega}{2k\Omega}\right)(a) = -2 \left(-8 \times 10^{-3} \times \frac{dx}{dt}\right)$$

$$\hookrightarrow y = 16 \times 10^{-3} \times \frac{dx}{dt}$$